



Department of Computer Science and Engineering
Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE 'B'

Course Code & Title: CS3353 & C Programming and Data Structures

Name of the Faculty member (s): Mrs. G. Mareeswari, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit IV / Hashing
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO10
- **Activity Chosen:** Jigsaw
- **Justification:**

Hashing is an important concept in data structures. Different types of hashing collision resolving techniques were assigned to students under each group. Each student in groups was given a chance to excel themselves in a particular technique and share that knowledge with their team members. The main theme of this Jigsaw activity is to promote both individual and peer teaching which requires students to understand the concept very clearly and engage in discussion and learning.

- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**

Based on the students' academic performance and abilities, they were divided into 8 groups of 8 students in each group. It is called homegroup. Each student in a group is assigned list of elements and a way to implement hash table. I delivered the instructions to the students on how to solve the problem using hashing resolution techniques. From each team one student is selected and framed as a team – “Expert Group”. 10 minutes were allotted to the teams to discuss the method of collision resolution and implementation of hash table. The expert group students were brought back into their home groups and allotted 15 minutes to explain the concept to other team members. Now each member in the team will be an expert in any one of hashing resolution techniques. They shared their ideas to all the other members in the group thus enabling each student to learn all the concept. This enabled the students to get insight knowledge in this topic.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO1	3	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	3	3	2	2	2	2
Justification for correlation	Requires basic engineering knowledge to implement hash table	Able to Identify engineering problems with hashing resolution techniques	Able to design solutions for complex engineering problems using hashing resolution technique	Able to solve collision in hashing individually or in team and implement hash table	Able to explain the way to resolve collision in hashing with their team members.	Able to apply hash data structures to solve various application in Information & communication Industry

- **Images / Screenshot of the practice:**



Figure 1: Students discussion with their team members

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Every student was actively participated and done the task and practice among their group members.
 - Each student explored the knowledge of the hashing and hashing resolution concepts among the other students in the class.
 - It enables students to learn thoroughly about hashing resolution methods.

- ❖ ***Benefit of the practice:***

- Students were actively participated in this activity.
 - From this activity, the students got clear idea about different hash resolution concepts.
 - Slow learners are also able to perform hash table implementation.

- ❖ ***Challenges faced in implementation:***

- It took some time and difficulties to organize the students into groups.
 - Some students in the team did not participate actively in the team.

References:

- ❖ <https://www.jigsaw.org/>
- ❖ <https://www.readingrockets.org/strategies/jigsaw>
- ❖ https://www.ritrjpm.ac.in/images/computer-science/36_IT6801_Jigsaw.pdf

Signature of Faculty Member

HOD